LeChal: Hi-tech Shoes That Show the Route

Paromik Chakraborty is a technical journalist at EFY

Technology is beautiful, especially when it comes to the aid of the differently abled. For instance, with LeChal, visually impaired people can travel anywhere without guidance from others. LeChal is basically a smart shoe that harnesses the visually challenged's heightened senses of touch. It enables them to travel by themselves with the help of vibrational patterns on their feet. This haptic-driven smart shoe has been produced by Secunderabad-based Ducere Technologies.

How a thought took shape

Back in 2012, Krispian Lawrence and Anirudh Sharma, two tinkerers with the common objective to help the visually challenged, came up with the idea of a smart shoe. The fact that the visually challenged are gifted with heightened senses of touch and hearing became their basis for designing something that could intuitively guide the user through gentle vibratory feedback by connecting to the directions set on a map. The thought process started in May 2012. In July of the same year, Sharma went ahead for further education, while Lawrence drove the project. The product was ready in 2016 and called 'LeChal.'

"LeChal as a concept came about as a solution to allowing one's senses to be utilised in the most efficient way. The thought led to identifying how to help the visually impaired to navigate in a way that their sense of touch (walking stick) or their sense of hearing (audio guidance) are not compromised," explains Lawrence. The first shipment of the product was made in July 2016. The four years in between were utilised for development, prototyping and upgradations to bring out what LeChal is today.

The LeChal package

Against all odds

Being a hardware product start-up, Ducere Technologies faced challenges different from software developers and service providers, in addition to the pressure of venturing into an unexplored field. The hardware ecosystem is not conducive for startups in India. So, putting together a team of people with compatible talent and ready to work for a startup project took much effort. Adding to the troubles was limited funding. The project was bootstrapped until it received angel funds. Despite all these hardships, the company has grown to become 120-employee strong today and drives sales through its own website as well as online and offline retail partners.

How LeChal works

Before getting into the basics of LeChal, let's first explain in simple words how haptic technologies work. Haptic technologies take data from a source and use tactile (touch-based) sensors to respond to it in the form of vibrations or touch-based sensations on the user. LeChal follows this basic principle to work.

LeChal has three iterations: the complete shoe form, a pair of insoles with a pair of sensor pods, and only the pods. The pods contain the main haptic sensors. These can either be buckled to the existing shoe-laces of the user or inserted into the insoles provided by LeChal that replace the existing insoles of the user's shoes. The sensors connect to the user's smartphone via Bluetooth. Users need to download the LeChal mobile application on their smartphone or tablet. The application provides a GPS (map) interface to set the desired destination via touchscreen or even via voice command, powered by the LeChal assist